

Why is the July-beam drift velocity low?

Ar/iC₄H₁₀ 90/10 contamination sims vs the run_58 data

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2026-07-20 (run_58 singles drift scan, nTOF EAR2, Ar/iso 90/10)

TL;DR: measured v_{drift} is $\sim 12\text{--}16\%$ below pure 90/10 Magboltz; Magboltz says only $\sim 0.2\%$ water explains it; air & O₂ are excluded — they would attach and kill the cathode-side charge, but the data amplitude is flat over the full 30 mm. $\Rightarrow$ trace water outgassing, not a leak.

The question

run58_scan/analyze_drift.py measures, on the clean Det A (nominal 30 mm gap), from micro-TPC track transit time $T_{\max}$:

- ▶ $v_{\text{drift}} = 35.6\text{--}35.8\text{ }\mu\text{m/ns}$ at $E = 200\text{--}233\text{ V/cm}$;
- ▶ pure 90/10 Magboltz predicts **40.5 / 42.6** $\Rightarrow$ **-12 to -16%**;
- ▶ and the curve *plateaus* where pure keeps climbing.

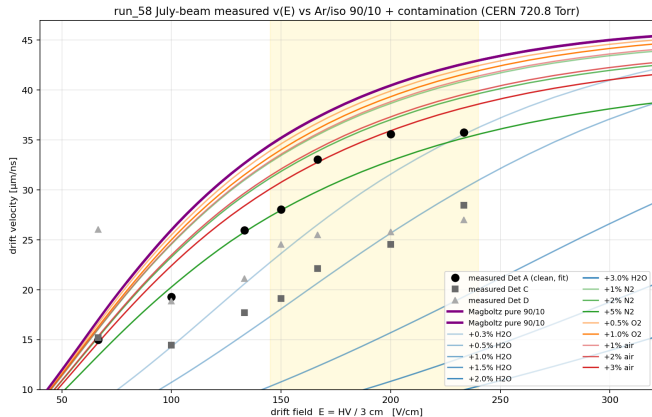
Long flush $\Rightarrow$ is the residual **water outgassing**, an **air leak**, or **O₂ permeation**?

(run_58 only had a bench 95/5 shape reference baked in — it flagged the need for a real 90/10 curve. This is it.)

$E\text{ [V/cm]}$	meas. A	pure 90/10
167	33.1	37.3
200	35.6	40.5
233	35.8	42.6

v in $\mu\text{m/ns}$, gap = 30 mm

The sim: 15 contaminated 90/10 mixtures (Magboltz, CERN 720.8 Torr)



Base Ar/iso 90/10 + H₂O (0.3–3%),
air (1–3%), O₂ (0.5–1%), N₂ (1–5%).
Run as 15 independent HTCondor jobs
on lxplus (LCG_107), each
 $v(E) + \eta(E) + \text{diffusion}$.

Det A (black) sits just below pure.

Water is a *potent suppressant*: even
0.5% halves v ; the **~1% that fit June
det3** would give $v \approx 16$, not 35.6.

On velocity *alone*, the deficit matches
~0.2% H₂O or ~3% N₂ or ~2–3% air
— **degenerate**.

The discriminator: attachment

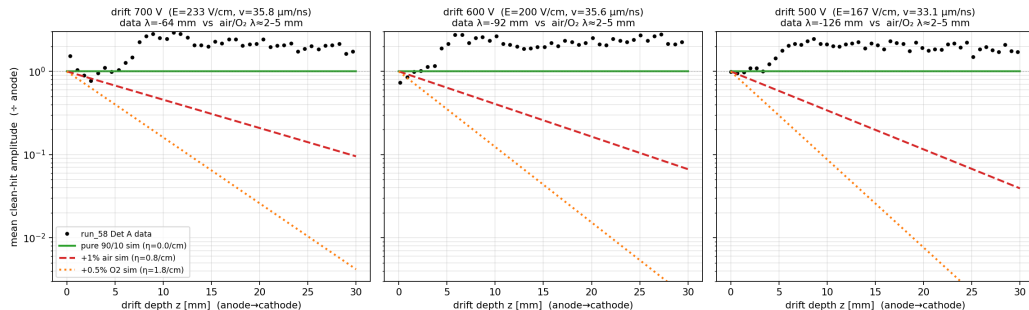
Any air/O₂ big enough to bend v down also **eats charge** over the drift.

hypothesis	fraction to match v	η @200 [cm ⁻¹]	charge surviving 30 mm
pure 90/10	—	0	100%
water (outgassing)	~0.2% H ₂ O	0	100%
N ₂ (inert)	~3%	0	100%
air in line	~2–3%	1.9–3.0	0.01–0.35%
O ₂ permeation	~1%	4.2	~0%

⇒ velocity is degenerate, but η is not. Two survivors with zero attachment: trace water or inert N₂ (which has no natural source — air ingress would drag O₂ in too). **Measure η on the data.**

The data settles it: amplitude vs drift depth (run_58 Det A)

run_58 Det A — clean-hit amplitude vs drift depth: DATA is flat (no attachment) → excludes air/O₂, consistent with trace water



Mean clean-hit amplitude vs depth (cached driftspec, time→depth via measured v). Data (black) is flat — $A_{\text{cathode}}/A_{\text{anode}} \approx 1.8-2.4$ across the full 30 mm; +1% air → 7%, +0.5% O₂ → 0.2% at the cathode. Data rejects air/O₂ by 1–3 orders of magnitude.

Conclusion

- ▶ The $\sim 12\%$ velocity deficit is **real** (confirmed vs a proper 90/10 CERN Magboltz curve).
- ▶ **Air-in-line and O_2 are experimentally excluded**: they would attach ($\lambda \approx 2\text{--}5\text{ mm}$) and obliterate the cathode-side charge — but the measured amplitude is **flat over the whole gap**.
- ▶ $\Rightarrow$ a no-attachment slow-down = **trace water ($\sim 0.2\%$, outgassing/permeation)**. **Exactly the worry, but mild** — and far drier than June det3 (which showed real attachment, $\lambda \sim 13\text{--}15\text{ mm}$).

Caveats / next:

- ▶ v_{meas} assumes exactly 30 mm; a smaller effective gap shrinks the deficit and the required water.
- ▶ Fold the 90/10 curve into run_58's effective-gap fit to pin gap & water fraction jointly.

Artifacts: `garfield.sim/mm_one_mixture.py`, `analyze_9010_contam.py`, `results/attachment_run58.py`, `CONTAM_9010_SUMMARY.md`.